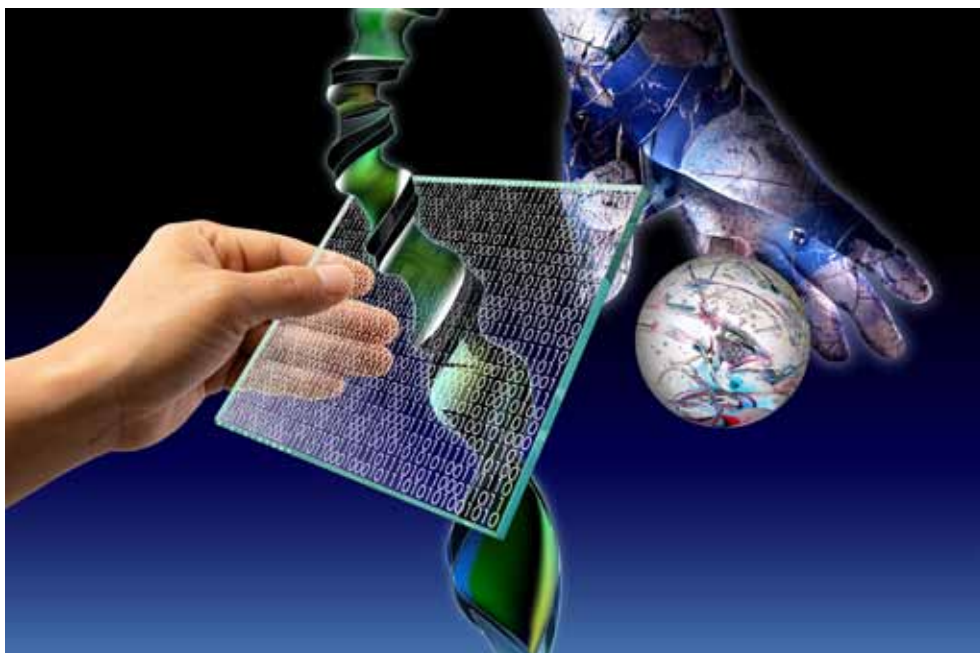


How A Glass Chip Can PRESERVE DATA FOR 10,000 YEARS

Imagine storing valuable information in something as simple as glass. It sounds futuristic, but researchers believe it could preserve data for thousands of years.



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In today's world, data has become one of our most valuable resources. Data is generated at almost every stage of modern life. As a result, preserving it securely and without degradation over long periods is not only a technical challenge but also an increasingly urgent necessity. Recently, a group of Microsoft researchers claimed that a glass sheet can preserve encrypted data for more than 10,000 years. On 18th February, *Nature* published the study. The researchers demonstrated that a sheet of borosilicate glass measuring 12cm wide and 2mm thick can store 4.8 terabytes of data, the equivalent of about 2 million books.

How did they achieve this?

Researchers at Microsoft used high-powered lasers to create tiny distortions within three-dimensional structures in borosilicate glass to embed hidden data. The encoded information could later be read using an optical microscope.

The idea of writing data onto glass or other durable media using lasers is not new. It is a maskless microfabrication technique in which complex three-dimensional nanostructures are created using high-energy focused laser pulses lasting femtoseconds (10^{-15} seconds). These tiny holes formed within the glass act as individual data storage units.

These storage units are known as voxels, the three-dimensional counterparts of pixels. One advantage of using lasers is that voxels can be packed densely by concentrating energy into an extremely small area. The storage capacity of each voxel can be increased further by adjusting the laser polarisation. Once written, the encoded data can be read using an optical microscope.

Why do scientists prefer glass?

Glass is better suited for long-term data storage because it can withstand changing temperatures, humidity, electromagnetic